

GAGANDEEP SINGH DHILLON

Data Scientist/Analyst

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SUMMARY

Aspiring **Data Scientist with 5+ years** of experience driving strategic insights through advanced data modeling, statistical analysis, and machine learning. Proven success in building predictive models, automating **ETL pipelines**, and **delivering** interactive dashboards for executive decision-making. Skilled in **MLOps** practices, cloud-based **deployments (AWS)**, and **integrating GenAI/LLMs** for enhanced analytics. Adept at cross-functional collaboration, with a strong understanding of credit risk, customer churn, and portfolio optimization.

PROFESSIONAL EXPERIENCE

American Express | New York, US
Data Scientist

Apr 2024 – Present

PROJECT OVERVIEW

Credit Risk Prediction and Management Platform: Developed a scalable predictive model using logistic regression and random forest algorithms to evaluate customer credit risk by analyzing transactional and demographic data. Using SQL and Python, performed detailed data analysis to identify key factors leading to credit defaults. Automated workflows with ETL pipelines managed by Apache Airflow and stored data on AWS Redshift for real-time analysis. Insights were presented to senior management through an interactive Tableau dashboard, enabling better decision-making. By implementing targeted interventions, reduced default rates by 12%, optimized risk strategies, and improved the overall health of the credit portfolio.

Responsibilities:

- Designed and implemented SQL-based data models to analyze millions of transactional records, uncovering untapped revenue opportunities and reducing unnecessary expenditure by 8%, leveraging AWS Redshift for optimized query performance and data warehousing.
- Pioneered the creation of advanced Tableau dashboards integrated with real-time data sources, facilitating instantaneous updates on KPIs and enabling faster decision-making by senior management.
- Applied predictive analytics using Python to develop a revenue forecast model (92% accuracy), streamlining budget allocation, and deployed it on AWS for scalable computation using **MLOps best practices (CI/CD pipelines with SageMaker & MLflow)**.
- Conducted a comprehensive study on credit portfolio risk using classification models (logistic regression, random forest), reducing default rates by 12% and enhancing portfolio stability.
- Engineered end-to-end ETL workflows to automate data integration across platforms, decreasing manual intervention and data processing time by 10%, utilizing Apache Airflow for workflow orchestration.
- Collaborated with marketing and strategy teams to design A/B testing frameworks, optimizing campaign effectiveness, and achieving a 15% improvement in conversion rates.
- Developed a churn prediction model using random forest and XGBoost, combined with **LLM-powered sentiment extraction** from customer feedback, retaining 5% of at-risk high-value clients.
- Explored **Generative AI models (OpenAI API)** for generating synthetic training data to handle class imbalance in predictive models, improving model robustness and reducing bias.
- Prepared executive-level presentations translating complex statistical findings into actionable insights, contributing to the approval of strategic initiatives worth over \$5M.
- Conducted trend analysis to identify key financial metrics influencing profitability, resulting in a 10% operational cost reduction through data-driven initiatives.
- Partnered with IT teams to implement robust data governance standards and enhanced pipeline observability, supporting compliance and **automated monitoring in MLOps workflows**.

United Airlines | New Jersey, US
Data Analyst

Jan 2023 – March 2024

PROJECT OVERVIEW

Operational Intelligence & Insights Platform (OIIP): Led strategic, data-driven initiatives to optimize operations, enhance customer experience, and support revenue growth. Delivered actionable insights using advanced analytics, partnering with cross-functional teams to drive decision-making and performance improvements.

Responsibilities:

- Extracted, cleaned, and transformed large-scale structured datasets using **SQL, Python (Pandas, NumPy)**, and **Excel**, enabling robust analytics and data-driven decisions across operations and marketing.
- Built and maintained automated **Tableau dashboards** to monitor **flight performance, revenue trends**, and **customer service KPIs**, reducing reporting turnaround time by **30%**.
- Partnered with marketing and revenue management teams to develop **predictive models** for customer segmentation and demand forecasting, leading to a **15% increase** in marketing ROI.
- Engineered **ETL pipelines** using **Alteryx** and custom Python scripts, streamlining data ingestion and reducing manual intervention across analytics workflows.

- Conducted **A/B testing** and **multivariate experiments** to evaluate loyalty program effectiveness, resulting in an **18% boost** in customer engagement.
- Designed financial and operational models to analyze **route-level profitability** and **cost efficiency**, leading to optimized resource allocation and **\$2M in savings** annually.
- Performed **root cause analysis** on flight delays and operational bottlenecks using statistical techniques, contributing to a **22% reduction** in delays on domestic routes.
- Automated recurring performance reports and KPI alerts using Python, SQL, and Tableau, ensuring timely dissemination of key business insights to stakeholders.
- Collaborated with data engineering teams to integrate **external APIs** (e.g., weather, fuel data) and enhance the breadth and accuracy of operational analytics.
- Implemented data validation and **data governance checks**, ensuring **data integrity**, security compliance, and audit readiness across analytics pipelines.
- Supported **machine learning model development** for customer churn prediction and satisfaction analysis using **scikit-learn**, **logistic regression**, and classification techniques.
- Worked with pricing analysts to apply **elasticity modeling** and optimize ticket pricing strategies, improving load factors while maintaining revenue targets.
- Presented analytical findings and data visualizations to senior leadership using storytelling techniques to inform **quarterly planning** and **strategic decisions**.
- Mentored junior analysts in **SQL optimization**, **dashboard design**, and **airline metrics interpretation**, fostering knowledge sharing and team development.
- Authored and maintained detailed documentation including data dictionaries, analytics process flows, and dashboard usage guidelines for **self-service BI**.
- Led data literacy initiatives by conducting internal workshops and training sessions, enabling over **40 business users** to adopt best practices in data analysis and visualization.

eClerx | Mumbai, MH

Jan 2019 – Dec 2021

Financial Analyst

PROJECT OVERVIEW

Automated Trade Reconciliation System: I created an automated trade reconciliation process for equities, derivatives, and fixed-income portfolios using Excel macros, VBA scripting, and SQL-based validation rules. This helped identify and resolve recurring discrepancies, improving data accuracy. I also worked with cross-functional teams to design and implement workflow improvements, reducing manual tasks and streamlining the trade lifecycle. Using Python, I developed custom monitoring reports to provide real-time insights. These efforts led to a 25% reduction in reconciliation errors and a 40% boost in processing efficiency, ensuring compliance with industry standards and faster settlement times.

Responsibilities:

- Directed the reconciliation of trade positions across **equities, derivatives, and fixed-income portfolios**, deploying **SQL-based validation rules** to enhance data accuracy.
- Automated complex trade confirmation processes using **VBA macros**, **cutting manual processing efforts by 30% and achieving greater consistency in reporting**.
- Coordinated cross-departmental efforts to address cash flow mismatches, using **advanced Excel functions** to track transactions and improve settlement timeliness by 15%.
- Identified recurring anomalies in pre-settlement trade workflows by **leveraging data analytics**, reducing unresolved trade discrepancies by 25%.
- Designed bespoke trade activity reports using **Python scripting**, reducing manual reporting time by 40% while ensuring compliance with audit requirements.
- Led a quality assurance initiative to revamp corporate action processing, introducing standardized templates and workflows that cut processing times by.
- Evaluated **loan syndication bottlenecks and proposed** an automated data tracking system, enhancing loan lifecycle efficiency by 18% while maintaining accuracy.
- Trained and mentored a team of junior analysts on trade lifecycle operations and advanced **Excel techniques**, **fostering team efficiency** and knowledge sharing.
- Created detailed variance and risk analysis reports for **high-priority clients**, **utilizing Monte Carlo simulations** to support informed decision-making during volatile market conditions.
- Collaborated with IT and operations teams to upgrade **trade reconciliation systems**, **incorporating client feedback** that improved process transparency and enhanced client trust.

EDUCATION

STEVENS INSTITUTE OF TECHNOLOGY

MS Finance (General), STEM (GPA 3.147)

Hoboken, NJ

BHARATI VIDYAPEETH DEEMED UNIVERSITY

Bachelor of Business Administration (Financial Management) (GPA 9.5/10)

Pune, MH, India

TECHNICAL SKILLS

Programming & Analytics:

SQL, Python (NumPy, Pandas, scikit-learn, Statsmodels, Matplotlib, Seaborn), R, Advanced Excel (PivotTables, VBA, Power Query), Version Control (Git, GitHub)

Data Visualization:

Tableau, Power BI

Machine Learning

Logistic Regression, Random Forest, XGBoost, Forecasting, NLP, A/B Testing

Generative AI & LLMs

OpenAI API, Amazon Bedrock, LangChain (basic), Synthetic Data Generation

MLOps & Deployment

MLflow, Docker, GitHub Actions, AWS SageMaker, CI/CD, Model Monitoring Workflow

Collaboration Tools:

JIRA, Agile Methodologies

Business Intelligence & Reporting:

Financial Dashboards, A/B Testing, Key Metrics Analysis

Financial Modeling:

Budgeting, Forecasting, Variance Analysis

Equities & Fixed Income:

Trade Reconciliation, Portfolio Optimization

Risk Analysis:

Credit Risk, Market Volatility, Value at Risk (VaR)

Investment Research:

Derivatives Strategies, Fixed Income Analysis, Discounted Cash Flow (DCF) Valuation
Planning to pursue CFA certification.

OTHER NOTABLE PROJECTS

• ECONOMETRIC ANALYSIS OF SOCIO-ECONOMIC FACTORS INFLUENCING CRIME RATES (ACADEMIC):

What did I do:

In this project, I analyzed socio-economic data from major U.S. cities using advanced STATA techniques, such as fixed and random effects models, to identify key factors influencing crime rates. I found significant links between income inequality, education levels, and crime, providing data-backed recommendations for policy changes. These insights were shared with policymakers, emphasizing how socio-economic disparities impact crime trends and guiding the development of community-focused solutions.

• TIME-SERIES ANALYSIS AND VOLATILITY MODELING OF STOCK PRICES (ACADEMIC):

What did I do:

In this project, I conducted a detailed time-series analysis of stock price movements using R, applying ARIMA and GARCH models to study volatility patterns and predict future prices. I calculated Value at Risk (VaR) to measure portfolio risks, helping stakeholders reduce exposure during market fluctuations. Through thorough back-testing and validation, I improved the accuracy of the models. I then shared a comprehensive risk assessment and forecasting report, offering investors actionable insights to make informed decisions in volatile markets.

CERTIFICATIONS

- SIE (FINRA)
- Fellow of Insurance Institute of India (FIII)
- Statistics & R (HarvardX on edX)
- CFA L-1 Candidate
- Finance & Quantitative Modeling Specialization
- Fintech Specialization Bloomberg Market Concepts

REFERENCES AVAILABLE UPON REQUEST